

Overview of the Code

This program implements the **ANDNOT logic gate** using the **McCulloch-Pitts Neural Network Model**.

The program:

1. Takes binary input combinations.
2. Uses fixed weights and threshold.
3. Calculates activation value (Yin).
4. Applies threshold logic.
5. Produces binary output (0 or 1).

The network works according to:

- Weight $w_1 = 1$
- Weight $w_2 = -1$
- Threshold $\theta = 1$

The output becomes:

- 1 only when $X_1 = 1$ and $X_2 = 0$
- Otherwise output is 0

Final Output:

0100

Line by Line Explanation

1. Define Input Combinations

```
inputs = [  
    [1, 1],  
    [1, 0],  
    [0, 1],  
    [0, 0]  
]
```

Explanation:

This list contains all possible binary input combinations for two inputs:

- X_1
- X_2

Input Combinations:

X1 X2

1 1

1 0

X1 X2

0 1

0 0

Viva Answer:

This section stores all possible binary inputs for the ANDNOT gate.

2. Define Weights

$w1 = 1$

$w2 = -1$

Explanation:

Weights are assigned to inputs:

- Weight of X1 = 1
- Weight of X2 = -1

These weights decide the importance of inputs.

Viva Answer:

Weights are used to multiply the input values in neural networks.

3. Define Threshold

$\theta = 1$

Explanation:

Threshold decides whether neuron activates or not.

If:

- $Y_{in} \geq \theta \rightarrow \text{Output} = 1$
- $Y_{in} < \theta \rightarrow \text{Output} = 0$

Viva Answer:

Threshold is the minimum activation value required to produce output 1.

4. Print Weights and Threshold

```
print("w1 =", w1)
```

```
print("w2 =", w2)
```

```
print("Threshold =",  $\theta$ )
```

Explanation:

Displays:

- Weight 1

- Weight 2
 - Threshold value
-

5. Print Table Heading

```
print("\nX1 X2 Yin Output")
```

Explanation:

Prints heading for output table.

6. Loop Through Inputs

for x1, x2 in inputs:

Explanation:

Takes one input pair at a time from the list.

Example:

- First iteration → x1=1, x2=1
- Second iteration → x1=1, x2=0

Viva Answer:

The loop processes each input combination one by one.

7. Calculate Activation Value (Yin)

```
yin = (w1 * x1) + (w2 * x2)
```

Explanation:

Calculates weighted sum.

Formula:

$$Yin = w_1x_1 + w_2x_2$$

Example:

For:

- X1 = 1
- X2 = 0

Calculation:

$$\begin{aligned} Yin &= (1 \times 1) + (-1 \times 0) \\ &= 1 + 0 \\ &= 1 \end{aligned}$$

Viva Answer:

Yin is the weighted sum of inputs.

8. Apply Threshold Logic

if $y_{in} \geq \theta$:

$y = 1$

else:

$y = 0$

Explanation:

Checks condition:

- If Y_{in} is greater than or equal to threshold $\rightarrow$ output 1
- Otherwise $\rightarrow$ output 0

Viva Answer:

Threshold function decides whether neuron is activated or not.

9. Print Final Result

```
print(x1, " ", x2, " ", yin, " ", y)
```

Explanation:

Displays:

- Input X1
 - Input X2
 - Yin value
 - Final Output
-

Step-by-Step Working

Case 1:

X1 = 1, X2 = 1

Calculation:

$$Y_{in} = (1 \times 1) + (-1 \times 1) = 0$$

Since:

$$0 < 1$$

Output:

$$Y = 0$$

Case 2:

X1 = 1, X2 = 0

Calculation:

$$Y_{in} = (1 \times 1) + (-1 \times 0) = 1$$

Since:

$$1 \geq 1$$

Output:

$$Y = 1$$

Case 3:

$$X1 = 0, X2 = 1$$

Calculation:

$$Y_{in} = (1 \times 0) + (-1 \times 1) = -1$$

Since:

$$-1 < 1$$

Output:

$$Y = 0$$

Case 4:

$$X1 = 0, X2 = 0$$

Calculation:

$$Y_{in} = (1 \times 0) + (-1 \times 0) = 0$$

Since:

$$0 < 1$$

Output:

$$Y = 0$$

Final Output Table

X1 X2 Yin Output

1 1 0 0

1 0 1 1

0 1 -1 0

0 0 0 0

Final Output:

0100

Important Viva Questions

Q1. What is McCulloch-Pitts model?

It is the first artificial neuron model developed by Warren McCulloch and Walter Pitts.

Q2. Why is it called threshold logic?

Because output depends on threshold value.

Q3. What is Yin?

Yin is the weighted sum of inputs.

Q4. What are weights?

Weights determine importance of inputs.

Q5. What is threshold?

Threshold is minimum activation value required to activate neuron.

Q6. What is binary output?

Binary output means output can only be 0 or 1.

Q7. What is ANDNOT logic?

ANDNOT gives output 1 only when first input is 1 and second input is 0.

Q8. Why is w2 negative?

Negative weight acts like NOT operation.

Q9. What is the final output of ANDNOT gate?

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Conclusion

We successfully implemented the ANDNOT logic gate using the McCulloch-Pitts Neural Network model using fixed weights and threshold logic.

Aim: Generate ANDNOT function using McCulloch-Pitts neural net.

Objective: Learning to design MP models for logic functions like ANDNOT

Theory: McCulloch-Pitts neural model: The early model of an artificial neuron is introduced by Warren McCulloch and Walter Pitts in 1943. The McCulloch-Pitts neural model is also known as linear threshold gate. It is a neuron of a set of inputs and one output. The linear threshold gate simply classifies the set of inputs into two different classes. Thus the output is binary. Such a function can be described mathematically using the equations. The McCulloch-Pitts model of a neuron is simple yet has substantial computing potential. It also has a precise mathematical definition. However, this model is so simplistic that it only generates a binary output and also the weight and threshold values are fixed. The neural computing algorithm has diverse features for various applications. Thus, we need to obtain the neural model with more flexible computational features.

Important Theory – ANDNOT Function using McCulloch-Pitts Neural Net

McCulloch-Pitts Neural Model

The McCulloch-Pitts (MP) Neural Model was introduced by Warren McCulloch and Walter Pitts in 1943.

It is the first mathematical model of an artificial neuron.

The MP neuron works on:

- Binary inputs (0 or 1)
- Fixed weights
- Threshold value
- Binary output (0 or 1)

The neuron calculates a weighted sum of inputs called activation value (Y_{in}).

The output depends on whether Y_{in} reaches the threshold value or not.

Working of McCulloch-Pitts Neuron

The neuron:

1. Takes inputs.
2. Multiplies them with weights.
3. Calculates weighted sum.
4. Compares result with threshold.
5. Produces binary output.

Activation Formula

$$Y_{in} = w_1x_1 + w_2x_2$$

Where:

- x_1, x_2 = Inputs
 - w_1, w_2 = Weights
 - Y_{in} = Activation value
-

Threshold Function

The neuron output is decided using threshold logic.

Output Rule

If $Y_{in} \geq \theta \rightarrow Y = 1$

If $Y_{in} < \theta \rightarrow Y = 0$

Where:

- θ (theta) = Threshold value
 - Y = Output
-

ANDNOT Logic Function

ANDNOT is a logic gate where:

- Output is 1 only when:
 - $X_1 = 1$
 - $X_2 = 0$

Otherwise output is 0.

Truth Table of ANDNOT

X1 X2 Output

1 1 0

1 0 1

0 1 0

0 0 0

Final Output:

0100

Weights and Threshold Used

For ANDNOT function:

- $w_1 = 1$
- $w_2 = -1$
- $\theta = 1$

Formula Used

$$Y_{in} = (1 \times X_1) + (-1 \times X_2)$$

The negative weight acts as NOT operation.